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A MODEL FOR A GEOTHERMAL WELL IN A HYDRAULIC GROUNDWATER FLOW FOR GROUND SOURCE HEAT PUMP SYSTEMS

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ABSTRACT

Heat pumps are mechanical systems that provide heating to a space in the winter, and cooling in the summer. They are increasingly popular because the same system provides both cooling modes, depending on the direction of the cycle upon which they operate. For proper operation, the heat pump must be connected to a constant temperature thermal reservoir which in traditional systems is the ambient air. In ground source heat pumps however, subterranean ground water is used as the thermal reservoir. To access the subterranean groundwater, “geothermal” wells are drilled into the formation. Water from the building heating or cooling system is circulated through the wells thereby promoting heat exchange between the coolant water and the subterranean formation. The potential for higher efficiency heating and cooling has increased the utilization of ground source heating ventilating and air conditioning systems. In addition, their compatibility with a naturally occurring and stable thermal reservoir has increased their use in the design of sustainable or green buildings and man-made environments.

Groundwater flow affects the temperature response of thermal wells due to advection of heat by physical movement of groundwater through the aquifer. Research on this subject is scarce in the geothermal literature. This paper presents the derivation of an analytical solution for thermal dispersion by conduction and advection from hydraulic groundwater flow for a “geothermal” well. This analytical solution is validated against asymptotic analytical solutions.

The traditional constant linear heat source solution is dependent on the ground formation thermal properties; the most

dominant of which is the thermal conductivity. The results show that as hydraulic groundwater flow increases, the influence of the ground formation thermal conductivity on the temperature response of the well diminishes. The diminishing influence is evident in the Peclet number parameter; a comparison of thermal advection from hydraulic groundwater flow to thermal conduction by molecular diffusion.

Keywords: Thermal storage; Ground coupled heat pump; Geothermal well; Standing column well; Well field; Analytical model; Hydraulic Flow; Groundwater Flow

INTRODUCTION

The thermal wells studied in this paper operate by flowing a working fluid through the well that interacts thermally and hydrodynamically with the ground formation. Other thermal well types, such as closed loop ground heat exchangers, do not permit the working fluid to interact with the ground but follow the same basic thermal principles. Figure 1 shows the basic functionality of a thermal well. The physics of such a problem can be separated into different time regimes. The thermal response for early times relies on the thermal mass of the well fluid as the main storage mechanism for thermal energy. Once the well fluid is thermally saturated and the thermal profile of the well is formed in the axial direction, the thermal physics become dependent on the thermal and hydraulic properties of the ground formation.

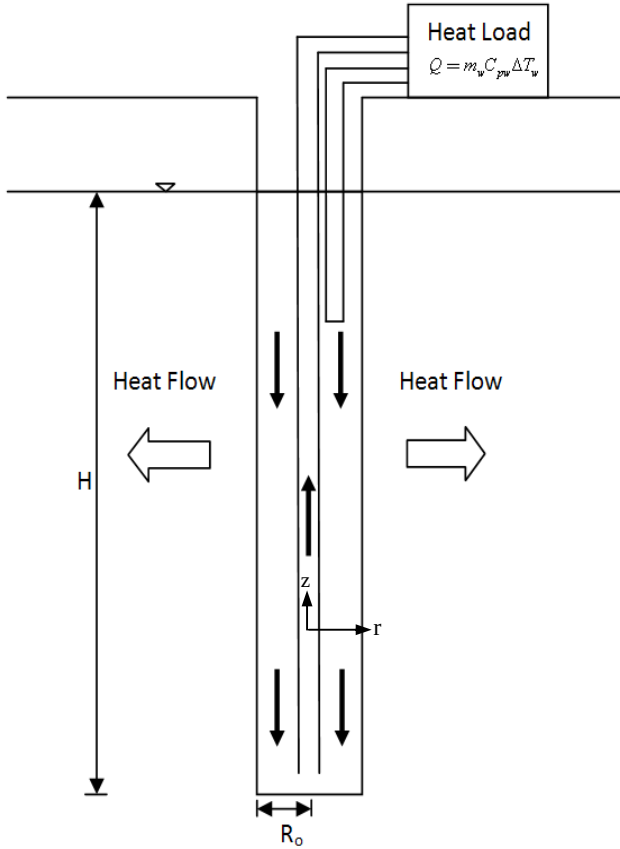


Fig. 1: General operation of a thermal well with negligible hydraulic flow. An external load is applied to the well through flow of a working fluid down an annulus and up a return pipe.

There are many analytical and numerical models used to simulate thermal wells, all with limitations and assumptions. Analytical models include the Kelvin line source model [1, 2] the cylindrical source model [3, 4], Eskilson's model using g-functions that incorporates the depth of the well [5] and the Finite line source model [6]. An in depth analysis of these analytical methods as well as some numerical methods is presented by Yang et al [7].

Yuill and Mikler [8] analyze the effect of induced groundwater flow in standing column wells due to pressure gradients between the well and the near-field ground formation due to water flow through the annulus. Claesson and Hellstrom [9] model a thermal well with ground water flow producing a g-function that takes Eskilson's conduction model [5] and superimposes the effect of groundwater flow using an infinite line source. Chaisson et al [10] reports results of a numerical model that analyzes the effect of groundwater flow on the thermal response of a borehole. These numerical and analytical models make the assumptions of homogeneous isotropic thermal and hydraulic ground formation properties but do not compare their results to asymptotic analytical solutions.

Gehlin and Hellstrom [11] show the effect of heterogeneous ground formation properties on the thermal response of a well given three scenarios: homogeneous ground

properties, a porous zone, an impermeable zone, and a well near a vertical fracture. Metzger et al [12] shows the effect of high flow on the thermal dispersion in the direction orthogonal to that of hydraulic flow for high Peclet numbers beyond those investigated in this paper and in a Reynolds number range that deviates from Darcy flow in ground formations, defined in Bear [13]. They use an infinite moving line source for the problem as is done in this paper but do not expand the solution to include the temperature variation around the circumference of the thermal well. Hecht-Mendez [14] presents a numerical analysis of a well with hydraulic flow for negligible hydraulic flow, a medium hydraulic flow where conduction is not negligible, and a high flow where conduction is negligible. Their results confirm that the numerical analysis software MT3DMS is able to solve heat transfer problems; however, the fundamental analytics incorporating the effect of the well geometry are not analyzed.

This paper presents the derivation of an analytical solution for the thermal response of a well under the influence of hydraulic flow normal to the axis of the well. Figure 2 shows a depiction of the problem.

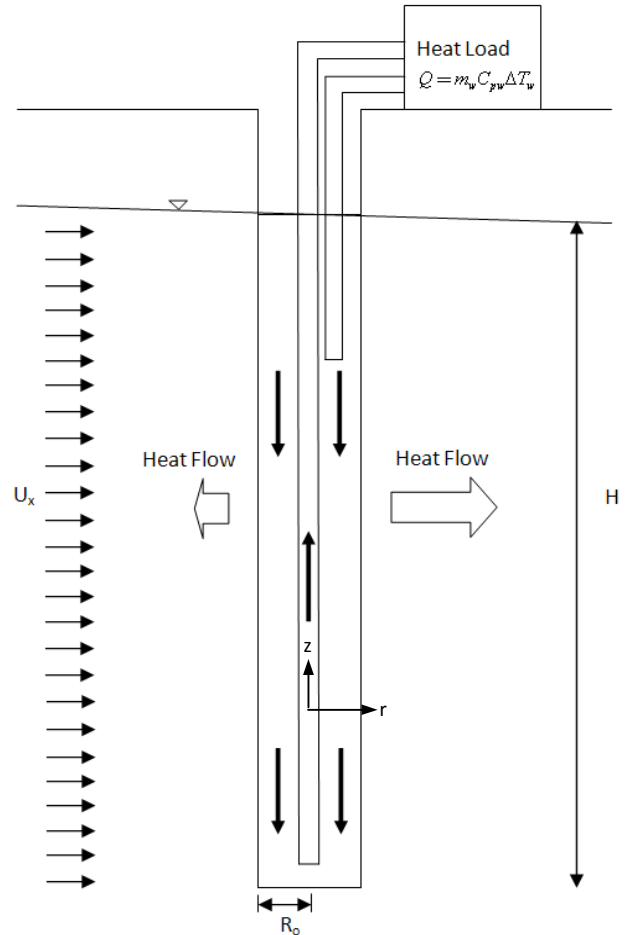


Fig. 2: Operation of a well with hydraulic groundwater flow and its effect on the dynamics of heat flow from the well.

The thermal physics of the problem deviate from conduction models due to the effects of advection of heat away from the well by the hydraulic flow. In pure radial conduction, the thermal profile grows infinitely with time due to the semi-infinite geometry in the radial direction. With hydraulic flow, the thermal profile plateaus due to the continual removal of heat from the well due to advection.

The analytical model begins with the solution for a continuous point source of heat with cross flow of fluid through the ground formation calculated by Carslaw and Jaeger [3] using Green's functions and a superposition method. This solution is integrated in a direction orthogonal to the path of flow producing a moving line source. An energy balance on the well integrates the moving line source solution around the circumference of the well to obtain the thermal well water temperature. This solution is validated against the solution for a stationary line source for early times, and the steady state solution for a moving line source for later times. The steady state solution for later times is analyzed to understand the influence of Nusselt number for axial flow through the well and Peclet number for hydraulic flow through the ground formation. Lastly, the analytical solution is used to produce contour plots for Peclet numbers of 0.01, and 1 for various times showing the effect of hydraulic groundwater flow.

ANALYSIS OF PROBLEM

In this section the thermal physics of the problem are described and analytical solutions are derived. To solve for the analytical solutions, the physics of the problem are identified in distinct time regimes. Early time solutions assume only heat transfer axially within the well, neglecting heat transfer to the ground formation. Later time solutions assume only heat transfer orthogonal to the axis of the well into the ground formation and neglect heat transfer in the axial direction. For early times, the thermal well undergoes a heating up phase where initially the thermal well temperature is equal to the infinite ground temperature, and at a later time, t^* , due to the addition of heat, the thermal well has achieved a steady temperature profile in the z -direction. The time, t^* , is equal to the residence time of flow through the thermal well. For later times ($t > t^*$), the thermal well predominantly transfers heat to the ground formation and the problem is analytically solved by assuming a constant heat load per unit length down the thermal well. The early time and later time solutions are superimposed to obtain the temperature profile of the thermal well water. The solutions are validated against asymptotic solutions.

Early Time Physics

Heat transported into the thermal well is absorbed by three mechanisms: the well thermal mass, the ground formation through conduction, and the ground formation through advection. The energy equation for flow down the well is given by

$$\rho_w C_{pw} V \frac{\partial T_w}{\partial t} + m_w C_{pw} H \frac{\partial T_w}{\partial z} + (hA)_e (T_w - T_i) + \rho_w C_{pw} U_x (2R_o H) \left(T_w - \frac{1}{\pi} \int_{\pi/2}^{3\pi/2} T_{wall}(\phi) d\phi \right) = 0 \quad (1)$$

Here $(hA)_e$ is the equivalent heat transfer resistance from the well to the stable infinite ground temperature and is a series combination of the convection resistance from the well to the wall of the well and the conduction resistance from the wall of the well to the infinite boundary where the temperature is T_i . Equation (1) scales to

$$\frac{\partial \theta_w}{\partial \tau_s} + \frac{\partial \theta_w}{\partial \zeta} + NTU(\tau_s) \theta_w + \frac{Pe_x}{Pe_z} \frac{2H}{\pi R_o} \left(\theta_w - \frac{1}{\pi} \int_{\pi/2}^{3\pi/2} \theta_{wall}(\phi, Pe_x, \tau_s) d\phi \right) = 0 \quad (2)$$

The scaling factor for temperature is $\theta = (T - T_i)/(Q/2\pi H k_g)$. θ_w denotes the well water scaled temperature, k_g is the thermal conductivity of the ground formation, and $\zeta = z/H$ is used to scale the z -direction. Scaled time is given by

$$\tau_s = \frac{m_w C_{pw} t}{\rho_w C_{pw} V} = \frac{t}{t^*} \quad (3)$$

τ_s is interpreted as the ratio of time to the residence time, t^* , of fluid flowing through the well.

$$Pe_i = \frac{\rho_w C_{pw} U_i R_o}{k_g} \quad (4)$$

$$NTU(\tau_s) = \frac{(hA)_e}{m_w C_{pw}} \quad (5)$$

An analytical solution for the thermal well temperature for early times ($t < t^*$), is obtained by assuming that the well thermal mass absorbs the majority of the energy input into the well. This assumes that the nonlinear third and fourth terms in Eq. (2) are negligible producing

$$\frac{\partial \theta_e}{\partial \tau_s} + \frac{\partial \theta_e}{\partial \zeta} = 0 \quad (6)$$

θ_e is used to denote early time scaled temperature. The solution to Eq. (6) is

$$\theta_e(\zeta, \tau_s) = \frac{2}{Pe_z} \frac{H}{R_o} \left(1 - \frac{\zeta}{\tau_s} \right), \quad \tau_s \leq 1, \quad 0 \leq \zeta \leq \tau_s \quad (7)$$

For typical systems, there are return pipes that channel fluid flowing down the well back to the top of the well to be used in an external application. This return flow interacts thermally with the annular flow down the well and its influence is solved analytically through superposition of a function, f_{sc} , transforming Eq. (7) into

$$\theta_e(\zeta, \tau_s) = \frac{2}{Pe_z} \frac{H}{R_o} \left(1 - \frac{\zeta}{\tau_s} \right) + f_{sc}(\zeta, \tau_s) \quad (8)$$

$f_{sc}(\zeta, \tau_s)$ is a function that is acquired from a global energy balance on the return pipe.

$$Q_{sc} = m_w C_{pw} \Delta T_{sc} = (UA)_{sc} \Delta T_{w \rightarrow r} \quad (9)$$

The maximum temperature variation between the annular flow and the return flow is at the top of the well and the minimum is zero at the bottom of the well. Assuming the linear profile in Eq.(8), the average temperature variation from the annulus flow to the return pipe flow is used for $\Delta T_{w \rightarrow r}$ and $f_{sc}(\zeta, \tau_s)$ is solved as

$$f_{sc}(\zeta, \tau_s) = \frac{\Delta T_{sc}}{\frac{Q}{2\pi H k_g}} = \frac{1}{Pe_z} \frac{H}{R_o} \frac{(UA)_{sc}}{m_w C_{pw}} \left(1 - \frac{\zeta}{\tau_s}\right), \quad (10)$$

$$\tau_s \leq 1, \quad 0 \leq \zeta \leq \tau_s$$

After $\tau_s > 1$, the thermal well temperature profile in the z-direction remains constant and is given by

$$\theta_e(\zeta) = \frac{2}{Pe_z} \frac{H}{R_o} (1 - \zeta) + \frac{1}{Pe_z} \frac{H}{R_o} \frac{(UA)_{sc}}{m_w C_{pw}} (1 - \zeta) \quad (11)$$

The early time temperature profile, that is only a function of the z-direction, is superposed with the later time solution derived in the following sections. The linear nature of the solution in this section allows for the assumption of constant heat load per unit time per unit length in the following sections. This is similar to the temperature profile from Woods and Ortega [15] where the temperature variation through the thermal well is dependent on ΔT_{HP} and ΔT_{SC} which are the first and second terms, respectively, in Eq. (11), in scaled form.

The validity of this analysis is dependent on the assumption made to obtain Eq. (6). The third and fourth terms of Eq. (1) are a comparison of heat transfer from conduction and advection, respectively, to the heat transfer due to axial flow down the well. Hence, as Pe_z increases, Eq. (11) becomes more accurate.

Late Time Physics: Pure Conduction Model Derivation

This section presents the solution for a pure conduction model where hydraulic flow is negligible. After $\tau_s = 1$, the well temperature profile is assumed to be thermally developed. The thermal physics transition to a ground formation conduction resistance dominated problem. This analysis assumes that the short term solution and long term solution can be superimposed to calculate the temperature profile in the ground formation and the well. The z-dependence is captured in the early time solution, and the r and ϕ -dependence are captured in the later time solution.

The pure conduction model is derived from the governing equation for radial conduction in a semi infinite solid at temperature T_i from an infinite line heat source. This governing equation is given as

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right) + \frac{\partial^2 T}{\partial z^2} = \frac{1}{\alpha_g} \frac{\partial T}{\partial t} \quad (12)$$

Here α_g is the thermal diffusivity of the ground formation. The second conduction term (z-direction) is negligible due to the

scaling parameter R_o^2/H^2 , which approaches zero. The form of the solution is reported by [3] as

$$T(r, t) - T_i = \frac{Q}{4\pi H k_g} \left(-\gamma + \ln \left(\frac{4\alpha_g t}{r^2} \right) - \sum_{n=1}^{\infty} \frac{1}{n \cdot n!} \left(-\frac{r^2}{4\alpha_g t} \right)^n \right) \quad (13)$$

Here γ is Euler's constant. Scaling Eq. (16) using $\tau = \alpha_g t / R_o^2$ and $\eta = r/R_o$ the solution simplifies for large values of τ to

$$\theta_L(\eta, \tau) = \frac{T(r, t) - T_i}{\frac{Q}{2\pi H k_g}} = \frac{1}{2} \ln \left(\frac{4}{\exp(\gamma) \eta^2} \tau \right) \quad (14)$$

Here θ_L denotes the scaled temperature for later times and is equivalent to the $\Delta T_G(t)$ term from Woods and Ortega [15] in scaled form. The well water temperature is calculated by including a convective resistance that incorporates the heat transfer from the well water to the well wall. This convective resistance is added to the conductive resistance in Eq. (14).

$$T_w - T_i = \frac{Q}{2\pi H k_g} \left(\theta_L(\eta = 1, \tau) + \frac{1}{Nu} \frac{k_g}{k_w} \right) \quad (15)$$

Typical Nusselt numbers from [16] for annular flow are of the order $10^2 - 10^3$ depending on the flow conditions. This leads to a negligible convective resistance as time becomes large.

Late Time Physics: Hydraulic Flow Model Derivation

The hydraulic flow equation in this section is first scaled using the governing equation and then solved using the solution for a moving point source that is integrated through the z-direction producing a moving line source. This solution is then modified to cylindrical coordinates (r, ϕ , z) and solved at $r = R_o$ producing the final form of the solution for the ground formation. The thermal well water temperature is then calculated through an energy balance on the well.

The governing equation for transient conduction with unidirectional hydraulic flow in Cartesian space assuming local thermal equilibrium is given by

$$(\rho C)_g \frac{\partial T}{\partial t} + (\rho C_p)_w U_x \frac{\partial T}{\partial x} = k_g \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right) \quad (16)$$

Here U_x is the volumetric flux of fluid through the formation also called the superficial velocity. When Eq. (16) is scaled, the governing equation becomes

$$\frac{\partial \theta}{\partial \tau} + Pe_x \frac{\partial \theta}{\partial \Phi} = \left(\frac{\partial^2 \theta}{\partial \Phi^2} + \frac{\partial^2 \theta}{\partial \Psi^2} \right) \quad (17)$$

Here $\Phi = x/R_o$, $\Psi = y/R_o$, and $\tau = \alpha_g t / R_o^2$. The Peclet number is defined as

$$Pe_x = \frac{(\rho C_p)_w U_x R_o}{(\rho C_p)_g \alpha_g} \quad (18)$$

This Peclet number is unique in that it incorporates the ratio of the heat capacities of the fluid flowing through the ground and the ground formation. Equation (17) is unsolvable given the

boundary conditions and an approximate method for solution is derived in the next section.

Moving Line Source Solution

Deriving an analytical solution for Eq. (16) is accomplished by assuming the thermal well to be a line source with constant heat load per unit length. To incorporate the effect of hydraulic flow through the thermal well, the line source is set into motion producing a thermal well flowing through the ground formation which, through a reference frame modification, is equivalent to the ground formation flowing through the thermal well. A final modification to the reference frame is made to incorporate the fact that only ground water, not the entire ground formation, is flowing through the thermal well.

The moving line source solution is derived beginning with equations from Carslaw and Jaeger [3]. The form of the governing equation for a stationary point source is

$$\frac{\partial T}{\partial t} = \frac{1}{\alpha_g} \left[\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} \right] \quad (19)$$

Here α_g is the thermal diffusivity of the infinite medium. Equation (19) is solved in [3] using Green's functions for an instantaneous point source; the point source is integrated with time to produce a continuous point source, and then modified to include a moving reference frame of $(x-U(t-t'), y, z)$ to solve for a moving continuous point source given by

$$T(x, y, z, t, U) - T_i = \frac{Q}{2(x^2 + y^2 + z^2)^{1/2} k_g \pi^{3/2}} \exp\left(\frac{Ux}{2\alpha_g}\right) \times \int_{\sigma_2}^{\infty} \exp\left(-\xi^2 - \left(\frac{U^2(x^2 + y^2 + z^2)}{16\alpha_g^2 \xi^2}\right)\right) d\xi \quad (20)$$

Here

$$\sigma_2 = \frac{(x^2 + y^2 + z^2)^{1/2}}{2\sqrt{\alpha_g t}} \quad (21)$$

Manipulating the moving point source into a moving line source, Eq. (20) is integrated through the z-direction to become

$$T(x, y, t, U) - T_i = \frac{Q}{2k_g \pi^{3/2}} \exp\left(\frac{Ux}{2\alpha_g}\right) \times \frac{1}{H} \int_{-\frac{1}{2}H}^{\frac{1}{2}H} \frac{1}{(x^2 + y^2 + z^2)^{1/2}} \int_{\sigma_2}^{\infty} \exp\left(-\xi^2 - \left(\frac{U^2(x^2 + y^2 + z^2)}{16\alpha_g^2 \xi^2}\right)\right) d\xi dz \quad (22)$$

Modifying Eq. (22) from Cartesian to cylindrical coordinates using $x = r \cos(\phi)$, $y = r \sin(\phi)$, and $r^2 = x^2 + y^2$, evaluating at $r = R_o$, scaling, and integrating through the z direction produces the moving continuous line source solution for all times.

$$\theta(\eta, \phi, Pe, \tau) = \exp\left(\eta \frac{Pe \cos(\phi)}{2}\right) \int_0^{\tau} \frac{\exp\left(-\frac{Pe^2 s}{4} - \frac{\eta^2}{4s}\right)}{2s} ds \quad (23)$$

Here $Pe = UR_o/\alpha_g$. The Peclet number is the ratio of advective thermal flow by fluid to thermal diffusion in the ground formation by solid and fluid. The advective thermal flow in the derivation above from [3], assumes that the infinite media is moving; however, for the hydraulic flow model, the working fluid in the ground formation is moving. Therefore, the reference frame used to obtain Eq. (20) is modified to $(x-(\rho C_p)_w/(\rho C_p)_g U_x(t-t'), y, z)$ for the current application where U_x is the volumetric flux of the working fluid through the ground formation producing $U = (\rho C_p)_w/(\rho C_p)_g U_x$. This modification is evident in Eq. (18). Hence, Pe from Eq. (23) is replaced by Pe_x defined in Eq. (18).

Solving ϕ Dependence

The temperature distribution in the ϕ direction at $r = R_o$ is non-uniform due to hydraulic flow of water through the ground formation. The well temperature is solved using the energy balance given in Eq. (1) and modifying the first two terms of the equation with a constant heat load (Q) evenly distributed along the well.

$$Q - h_o(2\pi R_o H) \left(T_w - \frac{1}{\pi} \int_0^{\pi} T(R_o, \phi) d\phi \right) - \rho_w C_{pw} U_x (2R_o H) \left(T_w - \frac{1}{\pi} \int_{\pi/2}^{3\pi/2} T(R_o, \phi) d\phi \right) = 0 \quad (24)$$

Solving for T_w and scaling produces

$$\theta_L(Pe_x, \tau) = \frac{\pi}{Pe_x + \pi Nu} + \frac{Pe_x}{Pe_x + \pi Nu} \left(\frac{1}{\pi} \int_{\pi/2}^{3\pi/2} \theta(\eta=1, \phi, Pe_x, \tau) d\phi \right) + \frac{\pi Nu}{Pe_x + \pi Nu} \left(\frac{1}{\pi} \int_0^{\pi} \theta(\eta=1, \phi, Pe_x, \tau) d\phi \right) \quad (25)$$

Since $U_z \gg U_x$ is valid for $Pe_x \ll \sim 100$, the effect of groundwater flow on the Nusselt number is neglected. The first term in Eq. (25) is neglected for $Pe_x > 10\pi Nu$, the second term is important for $\sim 0.1\pi Nu < Pe_x < \sim 10\pi Nu$, and the third term is important for $0 < Pe_x < \sim 10\pi Nu$. The minimum Nusselt number (obtained from [16]) for turbulent flow in an annulus is 8.5 but typical values are of the order $10^2 - 10^3$. The Peclet numbers in this analysis are small compared to the Nusselt number. Hence, the first and second terms in Eq. (25) are negligible for this analysis and produces

$$\theta_L(Pe_x, \tau) = I_0 \left(\frac{Pe_x}{2} \right) \int_0^{\tau} \frac{\exp\left(-\frac{Pe_x^2 s}{4} - \frac{1}{4s}\right)}{2s} ds \quad (26)$$

Figure 3 shows Eq. (26) varying Peclet number.

Carslaw and Jaeger [3] provide a function for the steady state temperature of Eq. (26) at large times for a moving line source. Their solution was obtained from integration of a steady state moving point source solution similar to the analysis for the moving line source solution. Scaling the equation from [3], modifying into cylindrical coordinates, evaluating at R_o , and integrating around the well produces

$$\theta_L(Pe_x) = K_0 \left(\frac{Pe_x}{2} \right) I_0 \left(\frac{Pe_x}{2} \right), \text{ for } 0 < Pe_x \leq 1 \quad (27)$$

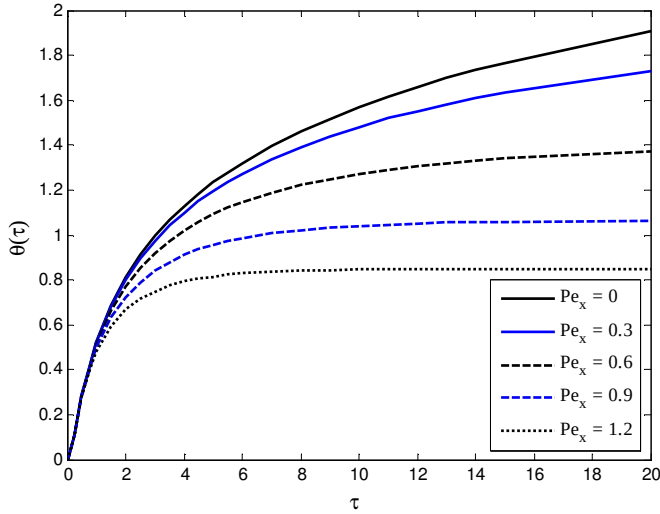


Fig. 3: Effect of varying Peclet number on the thermal response of the well.

By expanding the Bessel functions, Eq. (27) is simplified to

$$\theta_L(Pe_x) = \frac{1}{Pe_x}, \text{ for } 1 < Pe_x < 0.1\pi Nu \quad (28)$$

Figure 4 shows Eq. (26) validated against a scaled Eq. (13) for early times and Eq. (27) for later times. The time at which Eq. (26) deviates from Eq. (13) becomes shorter as Peclet number increases. The simplified solution as Peclet number approaches infinity is calculated from Eq.(25) to be

$$\theta_L(Pe_x) = \frac{\pi}{Pe_x}, \text{ } 10\pi Nu < Pe_x < \infty \quad (29)$$

It should be noted that as the Peclet number becomes large, the assumptions of uniform axial heat load fails. However, as shown in Fig. 3, for $Pe_x \gg 1$, $\theta(\tau) \rightarrow 0$ and the usefulness of a transient analysis diminishes.

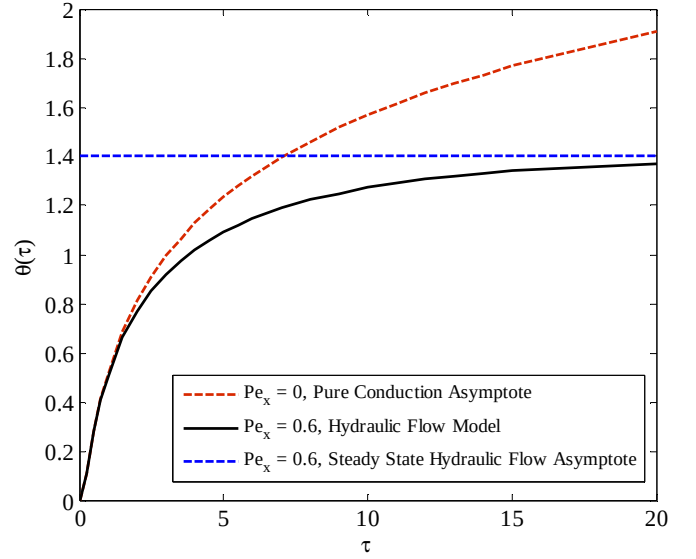


Fig. 4: Hydraulic flow model for $Pe_x = 0.6$ compared to the theoretical asymptotes for the solution given by the pure conduction model for early times and the asymptote given by the steady state hydraulic flow model for later times.

RESULTS AND DISCUSSION

In this section, the final analytical solutions for the thermal well temperature profile using the pure conduction model and hydraulic flow model are presented. These solutions superimpose the early time physics of the well with the later time physics of the ground formation given by

$$\theta_w(\zeta, Pe_x, \tau) = \theta_e(\zeta) + \theta_L(Pe_x, \tau) \quad (30)$$

This superposition method is the method used in [15] where $\theta_e(\zeta)$ is ΔT_{HP} and ΔT_{SC} in scaled form, and $\theta_L(Pe_x, \tau)$ is similar to $\Delta T_G(t)$ in scaled form with the addition of the hydraulic groundwater effects.

Results of the Hydraulic Flow Model

The analytical solution from Eq. (23) is plotted for $Pe_x = 0.01$, and 1 in space and time. These plots are shown in Figs. 5, and 6, respectively. When $Pe_x < 0.1$, the effects of thermal diffusion dominate the problem and the temperature contours are similar to the pure conduction model.

For $Pe_x = 0.1$ –10 a thermal plume migrating away from the well is seen due to the influence of the groundwater flow. For $Pe_x > 10$, the hydraulic groundwater advection completely dominates the problem and thermal diffusion is negligible.

For low Peclet numbers, the axisymmetric nature of the problem allows the line source to capture the problem accurately. However, for increasing Peclet number, as shown in Fig. 6, the infinitesimal size of the source limits its applicability in modeling a thermal well of finite size.

From these results, the application of the moving line source is limited to small Peclet numbers.

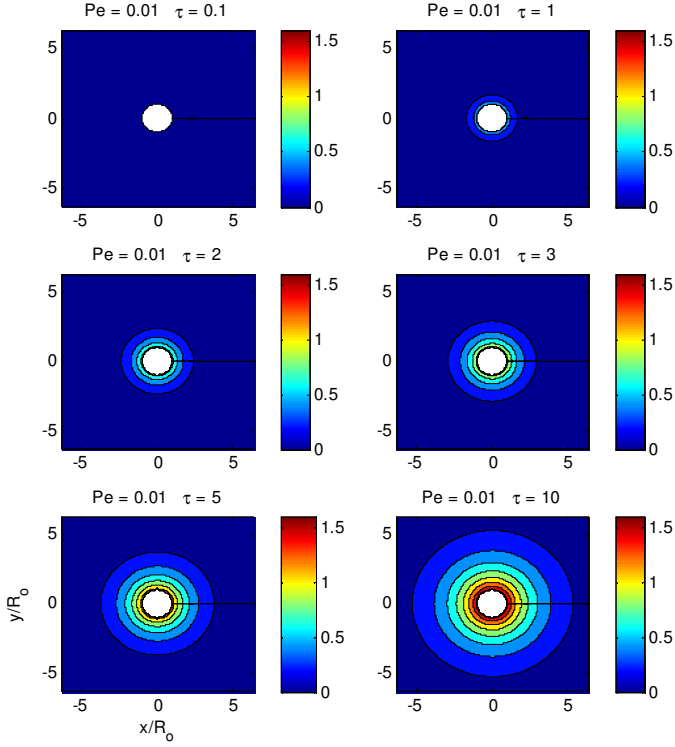


Fig. 5: Thermal profile in the ground formation for $Pe_x = 0.01$ at varying times.

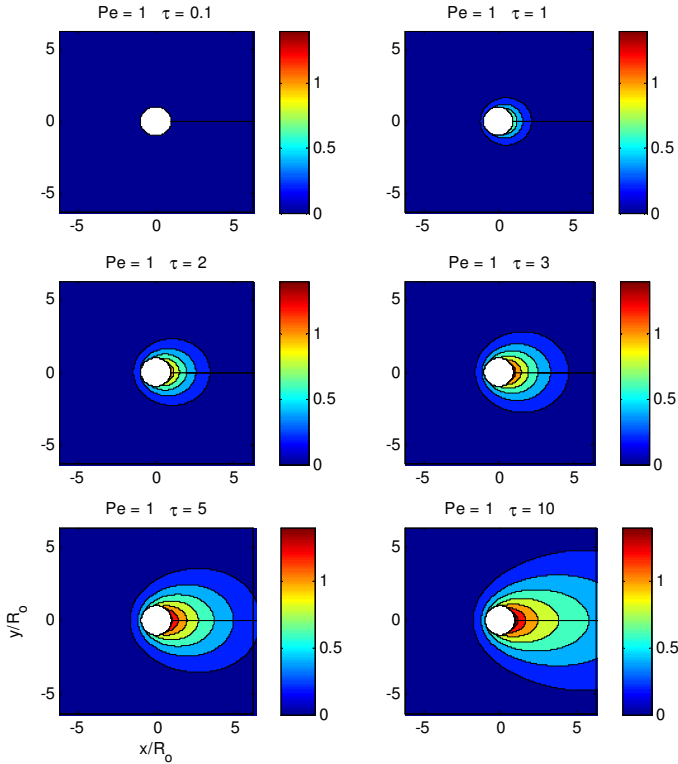


Fig. 6: Thermal profile in the ground formation for $Pe_x = 1$ at varying times.

Final Thermal Well Temperature Solutions

The pure conduction scaled thermal well water temperature solution is given by

$$\theta_w(\zeta, \tau) = \frac{2}{Pe_z} \frac{H}{R_o} \left(1 - \zeta + \frac{(UA)_{sc}}{2m_w C_{pw}} \zeta \right) + \frac{1}{2} \left(\ln \left(\frac{4\tau}{e^\gamma} \right) - \sum_{n=1}^{\infty} \frac{1}{n \cdot n!} \left(-\frac{1}{4\tau} \right)^n \right) \quad (31)$$

The hydraulic flow model solution for the well water temperature is given by

$$\theta_w(\zeta, Pe_x, \tau) = \frac{2}{Pe_z} \frac{H}{R_o} \left(1 - \zeta + \frac{(UA)_{sc}}{2m_w C_{pw}} \zeta \right) + \frac{1}{2} I_0 \left(\frac{Pe_x}{2} \right) \int_0^\tau \frac{\exp \left(-\frac{Pe_x^2 s}{4} - \frac{\eta^2}{4s} \right)}{s} ds \quad (32)$$

These solutions are limited by the validity constraints described in each section of the paper.

CONCLUSIONS

Analytical solutions for ground formations with hydraulic groundwater flow are poorly understood in the literature. This paper derived an analytical solution for a thermal well under the influence of hydraulic flow. The solution used a superposition method to separate the early and later time physics. The results were validated against asymptotic analytical solutions for the zero groundwater flow condition and the steady state hydraulic flow condition for infinite time. From the results it is concluded that

1. The early and later time governing equations are equivalent but are solved by neglecting different terms for each respective time regime, and the solutions are superposed to produce the final thermal well solution.
2. A transient solution exists for a thermal well under the influence of hydraulic flow orthogonal to the axis of the well and is obtained using the solution for a moving line source and an energy balance on the thermal well.
3. A steady state solution is obtained at shorter times as Peclet number increases and depends on both Peclet number and Nusselt number.

The limitations of this solution include:

1. The assumption that an infinitesimal sized heat source may be used to describe a finite sized thermal well.
2. The assumption of local thermal equilibrium in the ground formation which is a factor for ground formations composed of fractured rock.
3. The assumption of a homogeneous isotropic ground formation with homogeneous unidirectional groundwater flow.
4. The assumption of a uniform axial heat load condition which is valid only when Pe_z is large.

5. The assumption of uniform hydraulic water flow across the length of the thermal well.

For future consideration, a semi-analytical solution for the problem where the number of transfer units and ratio of Peclet numbers are not negligible will be calculated. This solution will divide the well into sections where, for each section, a linear temperature variation occurs and a uniform axial heat load is applied. The accuracy of the final semi-analytical solution will be dependent on the number of sections utilized.

The effect of multiple wells operating concurrently will be analyzed to understand the influence of a single well on another well. Also, the line source will be superimposed to simulate other geometric heat sources.

NOMENCLATURE

C_{pw}	Heat capacity of water (J/kgK)
H	Length of thermal well (m)
Nu	Nusselt number
Pe_x	Groundwater flow Peclet number
Pe_z	Annular well water flow Peclet number
Q_{sc}	Heat transferred from annular flow to return pipe flow (W)
R_o	Radius of thermal well (m)
T_i	Undisturbed ground formation temperature (K)
T_w	Thermal well water temperature
U	Reference frame velocity (m/s)
$(UA)_{sc}$	Overall heat transfer coefficient from annular flow to return pipe flow. (W/K)
V	Volume of water in thermal well (m ³)
k_w	Thermal conductivity of water (W/mK)
m_w	Mass flow rate of water through annulus (kg/s)
ρ_w	Density of water
$(\rho C_p)_g$	Volumetric heat capacity of formation (J/m ³ K)
$(\rho C_p)_w$	Volumetric heat capacity of water (J/m ³ K)
ΔT_{w-r}	Temperature difference between the annular water and return flow water in the thermal well. (K)
ΔT_{sc}	Temperature difference between the entrance to the return pipe and the exit of the return pipe. (K)

Subscripts and Superscripts

e	Early time regime parameter
g	Ground formation
i	Denoting a direction: x, y, or z
L	Later time regime parameter
o	Located at the outer wall of the thermal well
sc	Short circuiting of heat through return pipes
w	water in the thermal well

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